

12.11.2009

Adiabatna demagnetizacija kromovega galuna ($K_2SO_4 \cdot Cr_2(SO_4)_3 \cdot 24H_2O$)

→ paramagnet, Curiejev zakon

$$C_M = 1400 \frac{J}{kgK}$$

$$T' = 20^\circ C$$

↙ susceptibilnost

$$\mu(T') = 1,00029 \rightarrow \chi(T') = 2,9 \cdot 10^{-4}$$

$$\rho = 1830 \frac{kg}{m^3} = konst.$$

$$M: 0 \rightarrow 10^5 \frac{A}{m}$$

↑ izotermno

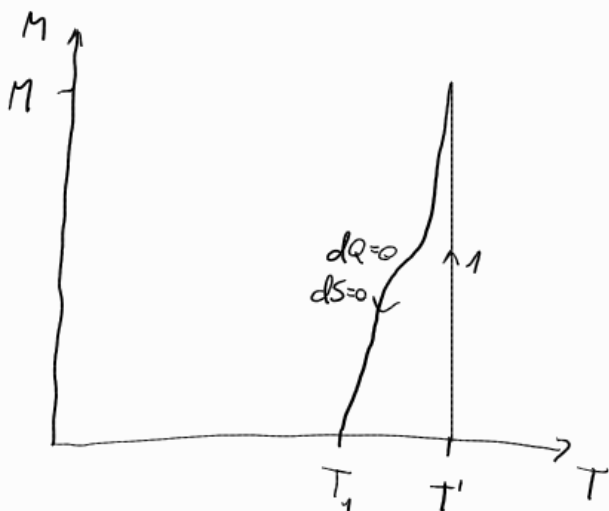
↓ adiabatno

$$T_m = -23^\circ C \text{ reverzibilno}$$

$$T_1 = ?$$

$$n = ? (\# \text{ ponovitev cikla, da je } T = T_m)$$

izberemo (M, H, T) sistem



$$0 = dS = \underbrace{\left(\frac{\partial S}{\partial T} \right)_M}_{\frac{mC_M}{T}} dT + \left(\frac{\partial S}{\partial M} \right)_T dM = 0$$

1. Maxwellova relacija

$$F = U - TS$$

$$dF = dU - TdS - SdT = \mu_0 V H dM - SdT$$

$$dU = TdS + \mu_0 V H dM$$

$$F = F(M, T)$$

$$dF = \left(\frac{\partial F}{\partial M} \right)_T dM + \left(\frac{\partial F}{\partial T} \right)_M dT$$

$$\left(\frac{\partial F}{\partial M} \right)_T = \mu_0 V H \rightarrow \left(\frac{\partial}{\partial M} \left(\frac{\partial \mu_0 V H}{\partial T} \right)_M \right)_T$$

$$\left(\frac{\partial F}{\partial T} \right)_M = -S \rightarrow \left(\frac{\partial}{\partial T} \left(\frac{\partial (-S)}{\partial M} \right)_M \right)_T$$

$$\left(\frac{\partial S}{\partial M} \right)_T = \mu_0 V \left(\frac{\partial H}{\partial T} \right)_M$$

2. Maxwellova relacija

$$G = U - TS - \mu_0 V M \mathcal{H} = F - \mu_0 V M \mathcal{H}$$

$$dG = dF - \mu_0 V (M d\mathcal{H} - \mathcal{H} dM) = \underbrace{-S dT}_{\text{red}} - \underbrace{\mu_0 V M d\mathcal{H}}_{\text{red}}$$

$$\boxed{+\left(\frac{\partial S}{\partial \mathcal{H}}\right)_T = +\mu_0 V \left(\frac{\partial M}{\partial T}\right)_{\mathcal{H}}}$$

$$\frac{m C_n}{T} dT - \mu_0 V \left(\frac{\partial \mathcal{H}}{\partial T}\right)_M dM = 0$$

Enačba stanja

$$M = \frac{C \mathcal{H}}{T} \quad (\text{Curie})$$

$$\left(\frac{\partial \mathcal{H}}{\partial T}\right)_M = \frac{M}{C}$$

$$\int_{T'}^{T_1} \frac{m C_n}{T} dT = \int_M^0 \mu_0 V \frac{M}{C} dM$$

$$C = \chi \cdot T' = 0,085 \text{ K}$$

$$9 C_n \ln \frac{T_1}{T'} = \frac{\mu_0}{C} \left(-\frac{M^2}{2}\right)$$

$$T_1 = T' \underbrace{e^{-\frac{\mu_0 M^2}{2 C C_n}}}_{q=0,974} = \underline{\underline{284,7 \text{ K}}} \rightarrow \underline{\underline{\Delta T = -8,3 \text{ K}}}$$

$$T_2 = T_1 q = T' q^2$$

$$\rightarrow T_n = T' q^n \rightarrow n = \frac{\ln \frac{T_m}{T_n}}{\ln q} < 5,4 \rightarrow \underline{\underline{\text{v } 6\text{-tem ciklu}}}$$

4.9) Jeklena palica

$$T = 20^\circ\text{C}$$

$$E_T = 2 \cdot 10^{11} \frac{\text{N}}{\text{m}^2}$$

$$\alpha = 10,6 \cdot 10^{-6} \text{ K}^{-1}$$

$$c_p = 500 \frac{\text{J}}{\text{kg K}}$$

$$\rho = 7800 \frac{\text{kg}}{\text{m}^3}$$

$$\epsilon_s = ?$$

$$\rightarrow (F, l, T) \text{ sistem} \rightarrow dl = \alpha l dT + \frac{l}{AE_T} dF$$

$$\downarrow$$
$$\epsilon_T = \frac{l}{A} \left(\frac{\partial F}{\partial l} \right)_T \rightarrow \epsilon_s = \frac{l}{A} \left(\frac{\partial F}{\partial l} \right)_s$$

$$\rightarrow S = S(F, l) \rightarrow dS = \dots dF + \dots dl \rightarrow \text{nerodine}$$

$$S = S(T, F) \rightarrow dS = \left(\frac{\partial S}{\partial T} \right)_F dT + \left(\frac{\partial S}{\partial F} \right)_T dF = 0$$

$$S = S(T, l) \rightarrow dS = \left(\frac{\partial S}{\partial T} \right)_l dT + \left(\frac{\partial S}{\partial l} \right)_T dl = 0$$

$$\left(\frac{\partial S}{\partial T} \right)_F = \frac{m c_F}{T}$$

$$dF = dU - T dS - S dT = F dl - S dT$$

$$dU = T dS + F dl$$

$$\left(\frac{\partial S}{\partial T} \right)_l = \frac{m c_l}{T}$$

$$\boxed{\left(\frac{\partial F}{\partial T} \right)_l = - \left(\frac{\partial S}{\partial l} \right)_T}$$

$$dG = dF - d(F l) = -S dT - l dF$$

$$\boxed{\left(\frac{\partial S}{\partial F} \right)_T = \left(\frac{\partial l}{\partial T} \right)_F}$$

$$\left(\frac{\partial S}{\partial F} \right)_T = \left(\frac{\partial l}{\partial T} \right)_F = \alpha$$

$$\rightarrow \frac{m c_F}{T} dT = - \alpha dF$$

$$\left(\frac{\partial S}{\partial l} \right)_T = - \left(\frac{\partial F}{\partial T} \right)_l = \alpha A E_T$$

$$\frac{m c_l}{T} dT = - \alpha A E_T dl$$

$$\frac{c_F}{c_l} = \frac{1}{AE_T} \left(\frac{\partial F}{\partial L} \right)_S \rightarrow \underline{E_S = E_T \left(\frac{c_F}{c_l} \right)}$$

$$c_p = c_F (F=0)$$

$$\underline{c_F - c_l = ?}$$

$F = \text{konst.}:$

hitreje: $S = S(T, L(T, F))$
 $c_F = \frac{1}{n} \left(\left(\frac{\partial S}{\partial T} \right)_L \left(\frac{\partial S}{\partial L} \right)_T \left(\frac{\partial L}{\partial T} \right)_F \right)$
 $\frac{1}{c_l} \quad \quad \quad \checkmark$

$$dS = \left(\frac{\partial S}{\partial T} \right)_F dT + \left(\frac{\partial S}{\partial F} \right)_T dF = \left(\frac{\partial S}{\partial T} \right)_F dT$$

$$dS = \left(\frac{\partial S}{\partial T} \right)_L dT + \left(\frac{\partial S}{\partial L} \right)_T dL = \left(\frac{\partial S}{\partial T} \right)_L dT + \left(\frac{\partial L}{\partial T} \right)_F dT + \left(\frac{\partial L}{\partial F} \right)_T dF$$

$$\left(\frac{\partial S}{\partial T} \right)_F dT = \left(\frac{\partial S}{\partial T} \right)_L dT + \left(\frac{\partial L}{\partial T} \right)_F dT$$

$$\frac{m c_F}{T} = \frac{m c_l}{T} + \left(\frac{\partial S}{\partial L} \right)_T \left(\frac{\partial L}{\partial T} \right)_F$$

$$\Delta C = c_F - c_l = \frac{1}{n} \left(\frac{\partial S}{\partial L} \right)_T \left(\frac{\partial L}{\partial T} \right)_F = \frac{1}{n} \left(- \frac{\partial F}{\partial T} \right)_L \left(\frac{\partial L}{\partial T} \right)_F = \frac{1}{n} \alpha A E_T (\alpha = \frac{1}{\rho} \alpha^2 E_T =$$

 $\underline{= 0,84 \frac{J}{kgK}})$

$$\rightarrow E_S = \frac{c_F}{c_l} E_T = \frac{c_F}{c_F + \Delta C} E_T = 2,0031 \cdot 10^{11} \frac{N}{m^2}$$

dodatek: $c_F(F) = ?$

$$\left(\frac{\partial c_F}{\partial F} \right)_T = \left(\frac{\partial}{\partial F} \left(\frac{1}{n} \left(\frac{\partial S}{\partial T} \right)_F \right) \right)_T = \frac{1}{n} \left(\frac{\partial}{\partial F} \left(\frac{\partial S}{\partial T} \right)_F \right)_T = \frac{1}{n} \left(\frac{\partial}{\partial T} \alpha \right)_F = \frac{1}{n} \alpha \cdot \alpha = \frac{T \alpha^2}{n} =$$

 $= \frac{T \alpha^2}{\rho A}$

$$\Delta c_F = \int_0^F \left(\frac{\partial c_F}{\partial F} \right)_T dF = \frac{T \alpha^2}{\rho A} F \rightarrow \frac{\Delta c_F}{c_F - c_l} = \dots = \frac{F}{A E_T} \ll 1$$

